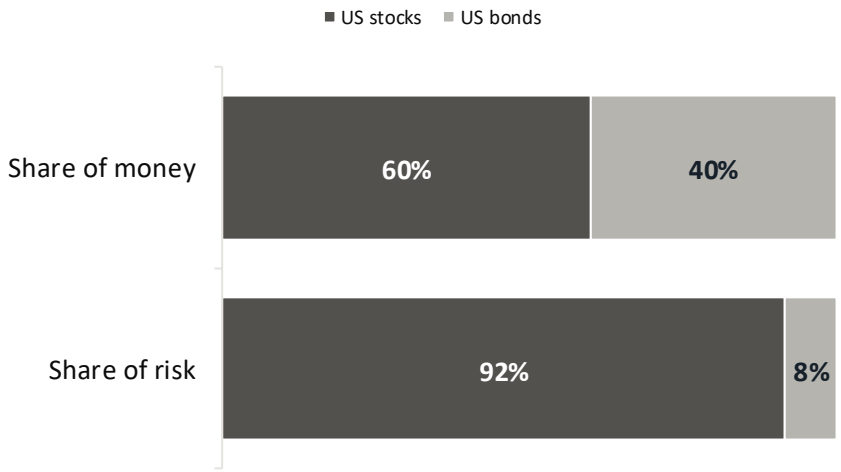
Sizing Private Markets in a 60/40

Private credit, private equity, real estate and infrastructure at 10%, 20% and 30%, with a 3-year lockup and a 15% liquidity floor

Selina Hach | October 2026

Inputs: J.P. Morgan 2026 Long-Term Capital Market Assumptions (10-15 year, USD, net of fees)

A 60/40 looks balanced, but 92% of its risk comes from stocks



When stocks fall, almost the whole portfolio falls with them. Bonds are 40% of the money but only 8% of the risk.

The question

Can moving 10%, 20% or 30% into private markets diversify that risk and improve return per unit of risk?

What I tested

Private credit, private equity, real estate and infrastructure.

The constraint

Private money is locked up for 3 years, and each strategy keeps 15% in cash for withdrawals.

The 20% lockup-safe portfolio improves return and reduces volatility

Compared with the baseline 60/40 portfolio

0.34 → 0.40

Sharpe ratio
Expected return: 6.73% → 7.06%
Volatility: 10.61% → 9.90%

20% fits

30% does not. Of the levels tested, 20% is the highest that meets the 3-year lockup limits.

Key assumption

Real estate provides the largest volatility reduction in the standalone tests, assuming a 0.35 correlation with stocks.

The 20% portfolio includes a 15% cash buffer within each private sleeve

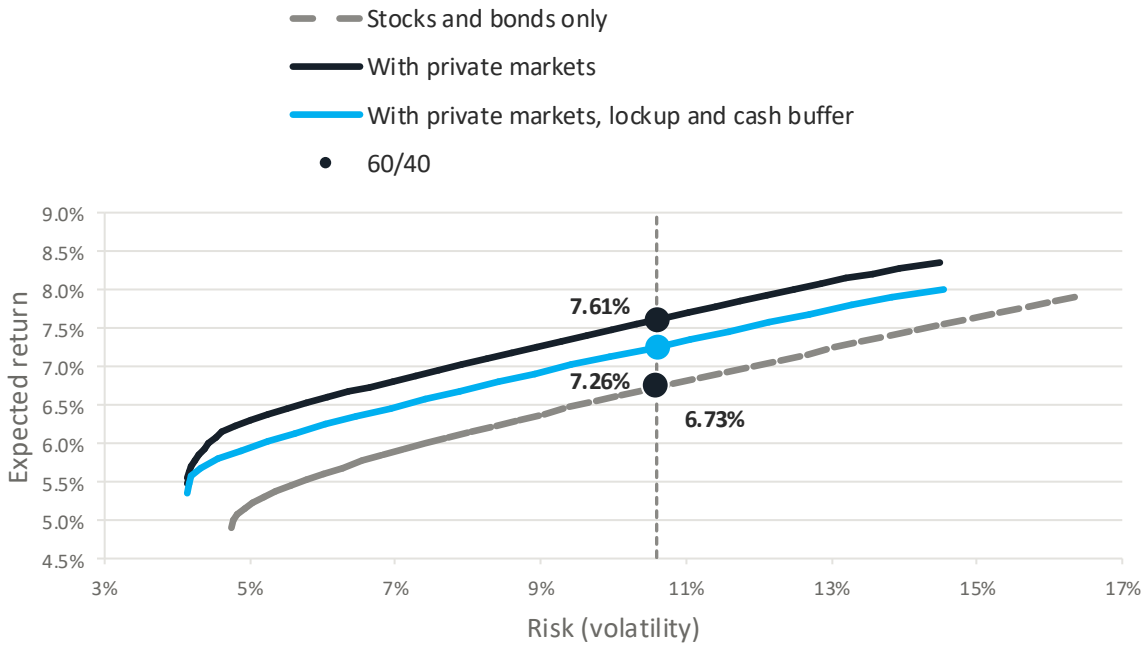
The modeled 20% allocation



Private: credit 8.7% · real estate 5.6% · private equity 4.7% · infrastructure 1.1%

At the 60/40's risk, privates add 53 bps after cash and lockup limits

Efficient frontier comparison at **10.61% volatility**



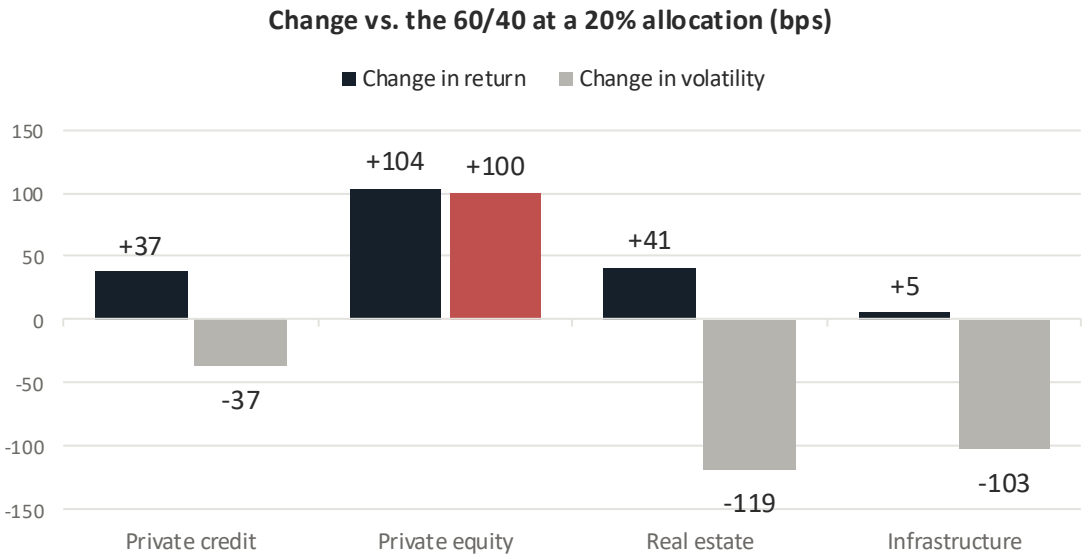
At the 60/40's risk	Expected return	Extra
Stocks and bonds only	6.73%	
With private markets	7.61%	+88 bps
With lockup and cash	7.26%	+53 bps

Each line shows the highest expected return available at each level of risk. Higher is better.

- Cash and lockup constraints reduce the expected return gain by 35 bps, leaving a 53 bps improvement over the 60/40.

Real estate cuts risk the most; private equity adds return but also risk

Each test allocates 20% to one private strategy, with 48% stocks and 32% bonds.

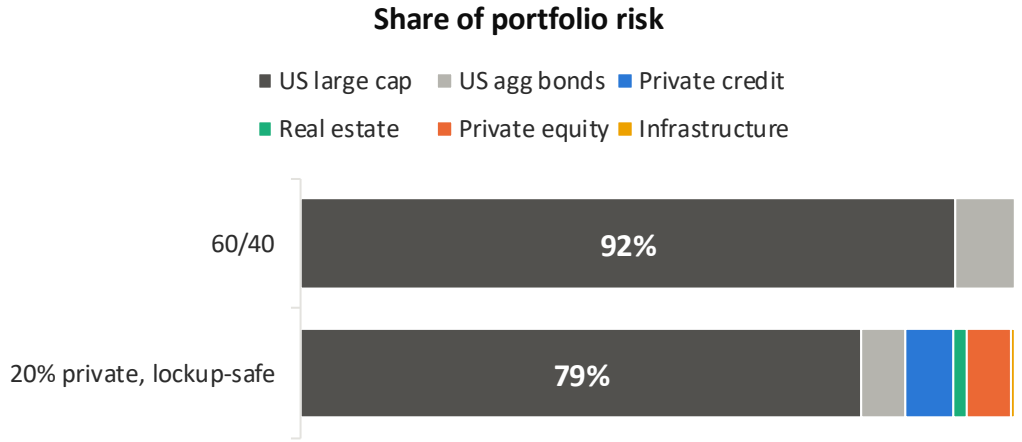


- **Real estate lowers risk the most** and still adds return. It moves least with stocks (*corr: 0.35*)
- **Private equity adds the most return**, but it is the only one that adds risk. It moves closely with stocks (0.75)
- Despite higher volatility, PE improves Sharpe by 0.060, the second-largest increase among the four standalone tests.
- Infrastructure lowers risk but adds almost no return (+5 bps)

At 20%	Private credit	Private equity	Real estate	Infrastructure
Change in Sharpe	+0.049	+0.060	+0.087	+0.042

Standalone tests exclude strategy caps, the cash buffer and lockup constraints.

The 20% lockup-safe portfolio reduces public stocks' risk to 79%



20% private, lockup-safe	Share of money	Share of risk
US large cap	48.0%	78.7%
US agg bonds	32.0%	6.1%
Private credit	8.7%	6.6%
Real estate	5.6%	2.0%
Private equity	4.7%	6.1%
Infrastructure	1.1%	0.5%

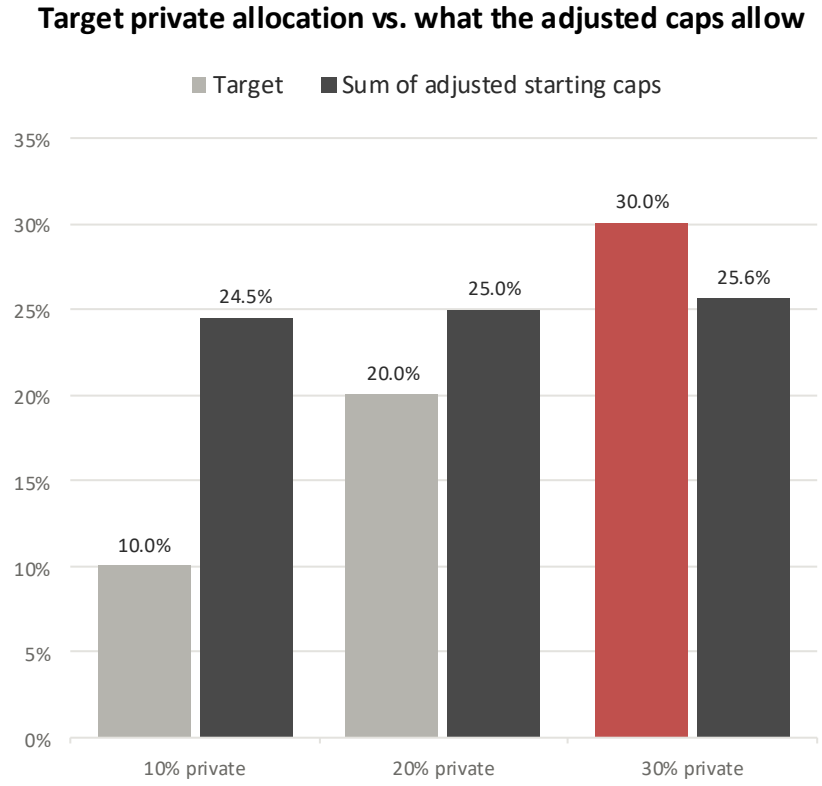
- In the 60/40, public stocks are 60% of the money but 92% of the risk.
- In the 20% lockup-safe portfolio, they drop to 48% of the money and 79% of the risk. Privates spread the risk out, but stocks are still the main driver.

These percentages show shares of total risk, not portfolio volatility.

Private equity accounts for 4.7% of capital but 6.1% of portfolio risk.

Real estate accounts for 5.6% of capital but only 2.0% of portfolio risk.

Under a 3-year lockup, 10% and 20% fit; 30% does not



Private holdings cannot be sold during the lockup. Their portfolio weights can rise as markets move, so starting below each cap leaves room for drift. Starting right at the cap, PE and real estate breach it in 86% to 89% of 10,000 simulated paths.

At 20% private	Original cap	Headroom	Adjusted cap	Actual weight	Breach prob.
Private credit	12%	1.38x	8.7%	8.7%	4.1%
Private equity	7%	1.50x	4.7%	4.7%	4.8%
Real estate	8%	1.44x	5.6%	5.6%	4.9%
Infrastructure	8%	1.33x	6.0%	1.1%	–
Total			**24.97%**	20%	

Headroom measures the simulated increase from starting weight to peak weight at the 95th percentile. *For real estate, an 8% cap ÷ 1.44 gives a 5.6% adjusted starting cap.*

Adjusted starting caps total 25.0% here and 25.6% at the 30% level, so 30% doesn't fit.

At 20%, **private equity, real estate and private credit reach their adjusted caps**. Infrastructure fills the remaining 1.1%.

Adjusted caps vary by scenario and target a breach probability of no more than 5% per sleeve.

20% is a workable private allocation under the modeled constraints

20%

Private allocation

+33 bps

Expected return

−71 bps

Volatility

Versus 60/40, after cash and lockup adjustments.

20% meets the modeled constraints; 30% does not.

Liquidity constraints change the allocation mix

20% private portfolio	Private credit	Real estate	Private equity	Infrastructure
Before adjustments	5.0%	8.0%	7.0%	0.0%
After (lockup-safe)	**8.7%**	**5.6%**	**4.7%**	**1.1%**

Lower starting limits for real estate and PE shift more of the allocation toward private credit.

TESTS BEFORE SELECTING A FINAL TARGET

Higher real estate–stock correlations · Allocations between 20% and 25%

Next step: turn one model portfolio into a tool for real clients

	1 Starting point	2 Constraints	3 Output
This model Built and tested	A clean 60/40 portfolio.	The same caps for everyone, a 15% cash buffer and a 3-year lockup.	One target mix: 20% private, tested against breach limits.
For a real client Next to build	The client's actual holdings, including private funds they already own.	Client-specific caps and liquidity needs. Rerun the optimization and lockup simulation.	A phased plan over several quarters that accounts for existing lockups, capital calls and distributions.

Reusable today: the optimizer, the lockup simulation and the Excel–Python tie-out. **New work:** client inputs and the timing of cash flows.

Appendix: capital market assumptions

10-15 year forecasts, USD. Private assets are median-manager, net of fees.

Asset	Expected return (compound)	Expected return (arithmetic)	Volatility	Sharpe ratio	Correlation with 60/40
US large cap	6.7%	7.9%	16.5%	0.29	0.99
US agg bonds	4.8%	4.9%	4.8%	0.38	0.46
Private credit	7.7%	8.6%	14.0%	0.39	0.52
Private equity	10.2%	11.9%	19.8%	0.45	0.70
Real estate	8.2%	8.8%	11.4%	0.50	0.30
Infrastructure	6.5%	7.0%	10.2%	0.38	0.45

Cash return 3.10% is used as the risk-free rate. I used forecasts rather than past data because private assets are only priced a few times a year, which makes their history look less risky than it is.

Appendix: full results by portfolio

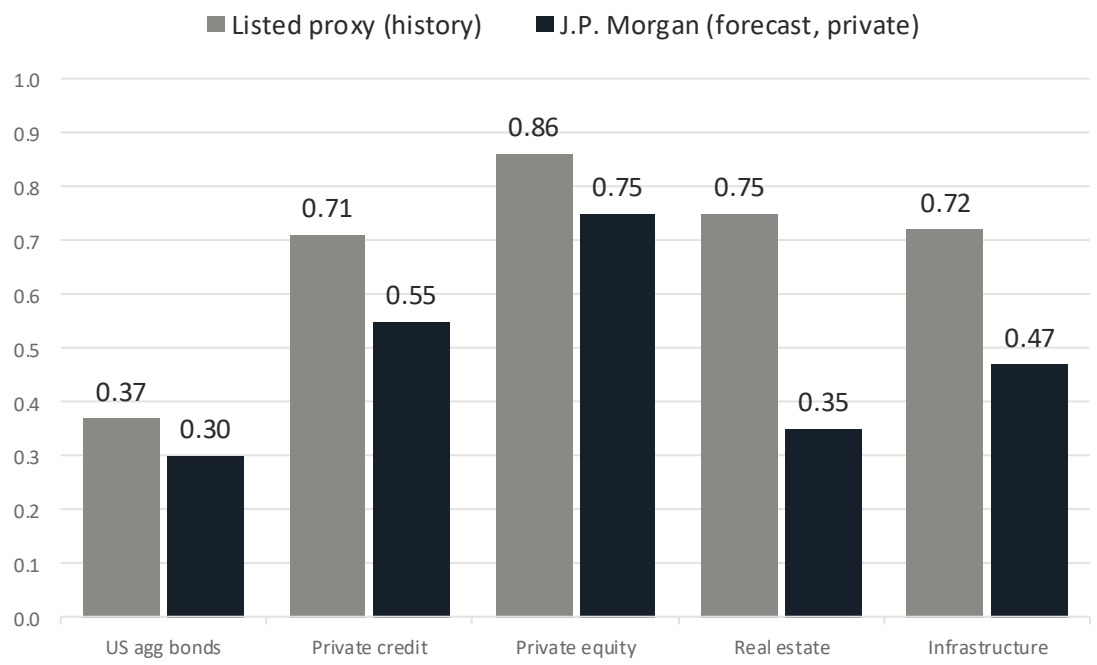
Highest-Sharpe split of the private allocation, each strategy under its cap. Total private size is fixed at 10%, 20% or 30%.

	60/40	10% private	20% private	30% private	10% lockup-safe	20% lockup-safe
US large cap	60.0%	54.0%	48.0%	42.0%	54.0%	48.0%
US agg bonds	40.0%	36.0%	32.0%	28.0%	36.0%	32.0%
Real estate	-	8.0%	8.0%	8.0%	5.4%	5.6%
Private equity	-	2.0%	7.0%	7.0%	4.6%	4.7%
Private credit	-	-	5.0%	9.3%	-	8.7%
Infrastructure	-	-	-	5.7%	-	1.1%
Expected return	6.73%	7.00%	7.35%	7.44%	6.97%	7.06%
Risk (volatility)	10.61%	10.14%	10.25%	9.92%	10.28%	9.90%
Sharpe ratio	0.34	0.38	0.41	0.44	0.38	0.40

Original caps: private credit 12%, real estate 8%, infrastructure 8%, private equity 7%. Lockup-safe columns keep 15% of each private holding in cash and start each one at or below its adjusted starting cap. 30% has no lockup-safe column because it doesn't fit.

Appendix: listed proxies move much more with stocks

Correlation with US large cap: monthly returns since 2015 vs. J.P. Morgan forecast



Strategy	Proxy
US agg bonds	AGG
Private credit	BIZD
Private equity	PSP
Real estate	VNQ
Infrastructure	IGF

Listed funds trade daily, so they pick up stock market swings.

Biggest sensitivity: if private real estate behaves more like REITs (0.75 vs. 0.35), it diversifies much less.

Appendix: method, assumptions and limits

How it works

- Inputs: J.P. Morgan 2026 expected returns, volatility and correlations.
- Private money comes out of stocks and bonds in proportion, so the public part stays 60/40.
- Python finds the split with the highest Sharpe ratio, each strategy under its cap.
- Lockup test: 10,000 simulated 3-year paths, quarterly steps. Stocks and bonds rebalance every quarter; private holdings can't be sold.
- Headroom = 95th-percentile peak weight ÷ starting weight. Adjusted starting cap = original cap ÷ headroom, then re-optimize until each sleeve breaches its original cap in at most 5% of paths.
- In the Excel file, caps are labeled "budget" and adjusted starting caps "effective budget".

Limits

- Median-manager returns: picking a good or bad manager matters a lot in private markets.
- Capital calls, distributions and existing private holdings are not modeled.
- Normal returns understate the risk of rare crashes.
- The caps, the 15% cash buffer and the 5% breach tolerance are my own assumptions.
- Results depend most on the real estate correlation assumption.
- The 15% cash buffer lowers each sleeve's return: PE 132, real estate 85, private credit 82, infrastructure 58 bps a year.